Running the priced inversion, and one self-check policy

`research/ic_acct4_inversion_20261021` · 2026-10-21 · class: accounting

Question

`bitop/v1` prices a field inversion as a binary extended Euclid and a quadratic-residue test as the same gcd-type loop. Each costs about 2.1 field multiplications at 6–16 bits.

The hyperelliptic arithmetic instead ran Fermat and Euler exponentiations through `num-bigint`'s windowed `modpow`. That is about 93 products whatever the size of p, so the work executed was some 40× what was charged.

Two questions follow:

- Does running the priced algorithms leave every committed count unchanged?
- Does it close the instruction gap the coverage diagnostic saw?

Method

- **The change.** Word-sized binary extended Euclid (`utils::inv_binary_u64`), Jacobi symbol (`utils::jacobi_u64`), a word path for `sqrt_mod_p` with the same roots, and square-and-multiply for small public powers. Charges per call are unchanged; prices and frozen vectors are untouched.
- **Equivalence.** Five rows of `hec_bitop_20261020` re-run with the study's seeds (5 IC and 40 rho runs each), compared field by field with the committed rows.
- **Diagnostic.** Callgrind with $(I, R) = (1, 2), (3, 2), (1, 6)$ trials. Each arm's figure is its marginal instructions over its marginal counted bitops.
- **Analysis.** `examples/ic_acct4_analyse.rs`.

Results

All five re-run rows (genus 2 at $p = 41, 61, 101$; genus 3 at $p = 23, 31$) are **byte-identical** to the committed study in per-run totals, phases, validation subsets, keys and verification.

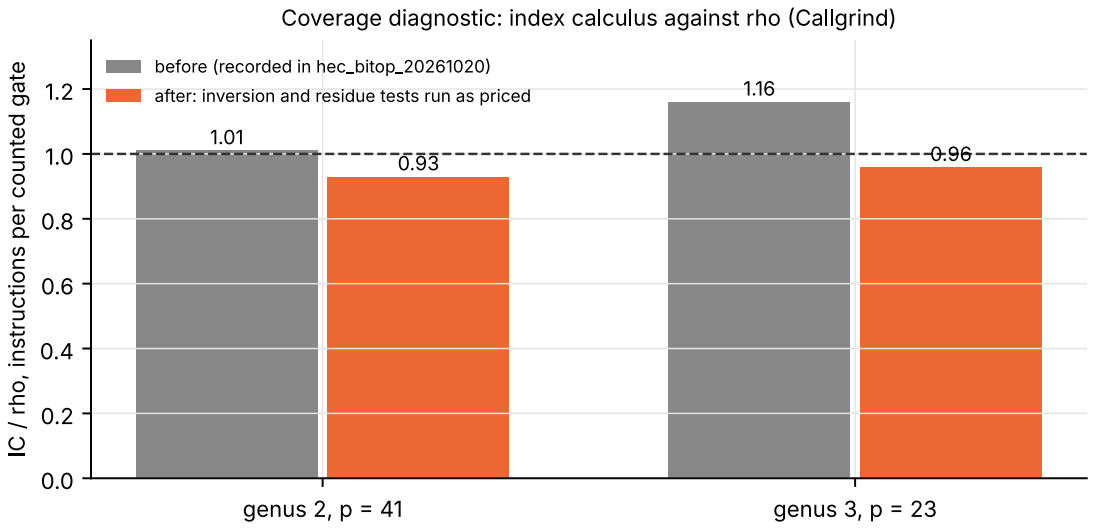


Figure 1: IC/rho instructions per counted gate. "Before" is the hec study's recorded whole-run figure; "after" is this study's marginal figure, a different method, so read the change as direction.

row	IC	rho	IC/rho now	before
genus 2, $p = 41$	1.212	1.304	0.930	1.01
genus 3, $p = 23$	1.611	1.681	0.958	1.16

Self-check policy

`docs/units/BITOPS.md` now has three rules:

- The domain is checked once, outside counted work.
- A method's check of its own answer is charged per target.
- Operand re-validation inside the arithmetic is counted but excluded from the headline.

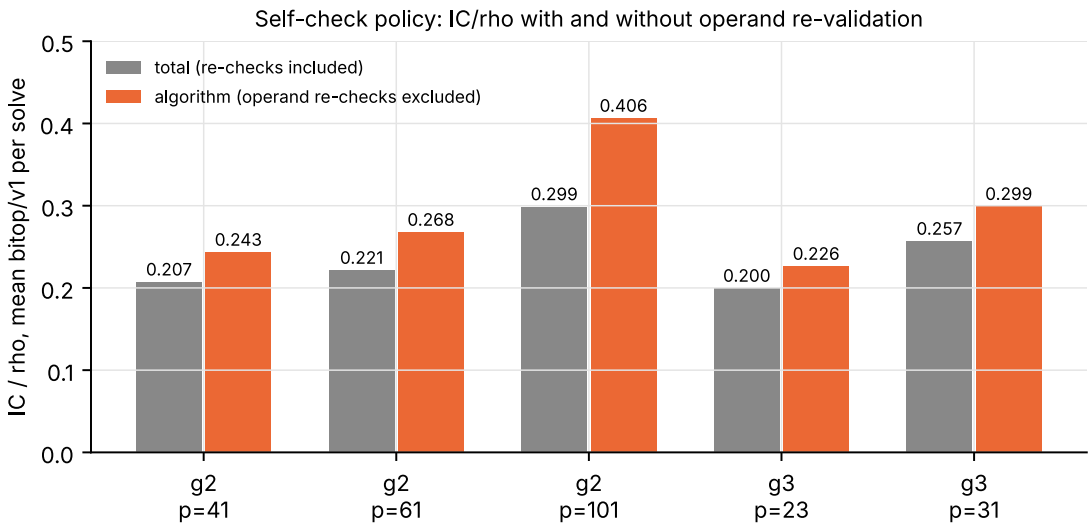


Figure 2: IC/rho per solve with and without the checked arithmetic's operand re-validation. Re-validation is a larger share of rho's work, so the ratio rises under the policy.

For the genus-4, $p = 61$ row, the policy-conforming figure is the study's existing "excluding validation" ratio, **0.132**, in place of 0.114. It is still below rho.

Scope

- Toy hyperelliptic Jacobians, counts only.
- Callgrind on one row per genus.
- No algorithm changed what it computes, so the class is accounting.